

Computational Physics Homework #1

Due 9/21

Submit via the Brightspace page for this class.

- 1) Differentiate the functions $\cos(x)$ and $\exp(x)$ at $x = 0.1, 10$ using single precision forward-, central- and extrapolated-difference algorithms.
 - a) Write a code that implements these three methods.
 - b) Make a log-log plot of the relative error ϵ vs step size h and check whether the scaling and the number of significant digits obtained agrees with simple estimates.
 - c) Truncation and roundoff error manifest themselves in different regimes in these plots. Clearly identify these regimes.

- 2) Consider the integral,

$$I = \int_0^1 \exp(-t) dt$$

And compare the relative error, ϵ , for the midpoint rule, trapezoid rule, and Simpson's rule for single precision.

- a) Write code that implements each method.
 - b) Make a log-log plot of ϵ as a function of number of bins N . (Since this is a log-log plot, choose values that scale in a reasonable fashion). Make N large enough such that you see the effects of roundoff error.
 - c) Explain what you see in the plot.
- 3) In cosmology, density fluctuations in the matter distribution are characterized by a power spectrum, $P(k)$, the rms amplitude fluctuations of the density waves, as a function of wavenumber k (with units of h/Mpc). In configuration space, these density fluctuations are described by the correlation function, $\xi(r)$, at a given scale r , usually in Mpc/h . These two quantities are related by

$$\xi(r) = \frac{1}{2\pi^2} \int dk k^2 P(k) \frac{\sin(kr)}{kr}$$

With this homework, I have attached a tabulated power spectrum. The first column is k and the second column is $P(k)$ (pay no attention to the third column). Using whatever integration method you prefer, use the above equation to calculate $\xi(r)$ in the range $r=[50,120]$ Mpc/h. The power spectrum is tabulated in logarithmic intervals in k , due to its power-law like nature. You may choose to use an interpolation technique, such as cubic spline, to help evaluate the integral.

Around $k \sim 0.1$, you can see oscillatory behavior in $P(k)$. We call these the “baryon wiggles,” and they manifest as a single “bump” in the correlation function at large scales. Using your calculation for $\xi(r)$, determine the scale, r , of the peak of this bump. Make a plot of $r^2\xi(r)$ over the required range in r (multiplying by r^2 visually enhances the bump). Indicate on this plot the scale of the peak, also known as the “baryon acoustic oscillation” (BAO) peak.

Notes:

- a) If you use spline interpolation, you are not required to code that up yourself. You may use a pre-packaged routine (or numerical recipes code).
- b) Formally, the limits of the integral are from $k=0$ to $k=\infty$. Note that $P(0)=0$. You may choose a finite upper limit, provided you can determine if your limit is robust.
- c) The ‘h’ in the distance units refers to the Hubble constant $h=H_0/100$, which sets the distance scale and is thus incorporated into the distance units, since its value is unknown.